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TENSIONING INSTRUMENTS

Abstract

Tensioner instruments, systems, and methods of tensioning flexible bands in-situ. The tensioner instrument may include first and second pivoting arms with a base for pinching and pulling the flexible band to provide tension. The tensioner instrument system may include a clip inserter and a ratcheting tensioner for tensioning the band through an implant. The instruments may be configured to apply and maintain tension to the flexible band, thereby providing the desired correction to the spine.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATION [0001] The present application is a continuation of U.S. patent application Ser. No. 18/057,317, filed on Nov. 21, 2022, which is a continuation of U.S. patent application Ser. No. 17/064,833, filed on Oct. 7, 2020 (published as U.S. Pat. Pub. No. 2022-0104859), all of which are incorporated herein by reference in their entirety for all purposes.

FIELD OF THE INVENTION

[0002] The present application relates generally to surgical instruments, and more particularly, tensioning instruments, for example, for spine surgery.

BACKGROUND OF THE INVENTION

[0003] Many types of spinal irregularities cause pain, limit range of motion, or injure the nervous system within the spinal column. These irregularities may result from, without limitations, trauma, tumor, disc degeneration, and disease. Often, these irregularities are treated by immobilizing a portion of the spine. This treatment typically involves affixing screws, hooks and/or clamps to one or more vertebrae and connecting the screws, hooks and/or clamps to an elongate spinal rod that stabilizes members of the spine.

[0004] Flexible bands may be used to achieve correction and provide fixation as an alternative and/or supplement to pedicle screws during spinal deformity surgery. The bands may be wrapped around bony anatomy and then a force may be applied to secure the spine to the spinal rod. Correction of the spinal deformity may be achieved and held by application of tension to the flexible band. There exists a need for improved tensioning instruments configured to apply tension to the bands.

SUMMARY OF THE INVENTION

[0005] To meet this and other needs, instruments, systems, and methods of tensioning flexible bands in-situ are provided. After a flexible band is wrapped around bony anatomy, such as a lamina or transverse process, the instruments may be configured to apply and maintain tension to the flexible bands, thereby providing the desired correction to the spine.

[0006] According to one embodiment, a tensioner instrument for tensioning a band may include first and second pivoting arms and a base. The first and second pivoting arms may each extend from a proximal end to a distal end. The first and second pivoting arms may each include handles near the proximal ends. The base may have a generally L-shaped body coupled to the first and second pivoting arms. The base may define a slot to retain the band and/or guide the distal end of the first pivoting arm. The second pivoting arm may define an opening located beneath the slot and sized and dimensioned to receive the band. When the handles are compressed together, the band may be pinched in the slot between the distal end of the first pivoting arm and the base, and a distance between the distal ends of the arms is increased, thereby applying a tension to the band. A ratchet may be positioned between the proximal ends of the first and second pivoting arms, thereby allowing for incremental tensioning.

[0007] The tensioner instrument may include one or more of the following features. The base may include a first elongate portion and a second elongate portion angled relative to the first elongate portion at a corner. The corner of the first and second elongate portions may couple with the first pivoting arm at a first hinge. The second elongate portion of the base may terminate at a free end, which couples with the second pivoting arm at a second hinge. The first hinge may move the first pivoting arm towards the base and the second hinge may move the base away from the second pivoting arm. The first end of the ratchet may be coupled to one of the first and second pivoting

arms via a pivot pin and the opposite end of the ratchet may be positionable through a slot in the other of the first and second pivoting arms. The ratchet may be a linear ratchet having a plurality of teeth along an interior of the ratchet. A pawl in the slot may be configured to engage the teeth to thereby incrementally maintain the position of the first and second pivoting arms and the amount of tension applied to the band.

[0008] According to one embodiment, a tensioner instrument may include a clip inserter and a ratcheting tensioner. The clip inserter may include a cannulated main body extending from a proximal end to a distal tip, a collar configured to translate along the main body, and a threaded shaft configured to move the collar. The ratcheting tensioner may include a fixed handle coupled to the clip inserter, a pivoting handle coupled to the fixed handle, a ratchet assembly including a pair of rotary ratchets and a spool keyed to the ratchets and positioned between the ratchets, and an actuator assembly including an actuator configured to engage the first and second rotary ratchets. When the pivoting handle is squeezed toward the fixed handle, the actuator may contact the ratchets and forces the ratchets and spool to rotate.

[0009] The tensioner instrument may include one or more of the following features. The collar may be a ring with two arms extending toward the distal tip of the main body. Each of the arms may define a notch configured to secure a spinal rod when the collar moves distally. The threaded shaft may define a hollow body such that a driver shaft is passable through the threaded shaft. The main body of the clip inserter may include a wire cut and one or more grooves on opposite sides of the wire cut configured to engage with pins on the collar. The actuator assembly may be positioned in the pivoting handle and include an actuator button for engaging and disengaging the actuator, an actuator pin for securing the actuator, and an actuator spring causing the actuator to contact the ratchets. The ratcheting tensioner may include a release assembly including a release arm pivotably coupled to the fixed handle with a pin and a release spring forcing the release arm into contact with the ratchets at rest. The release arm may include a body with a thumb press and a pair of spaced apart tongues for engaging the ratchets. The ratcheting tensioner may include a button assembly for securing the clip inserter to the ratcheting tensioner, the button assembly including a button, a button pin for securing the button, a stop pin for engaging with the body of the clip inserter, and a button spring causing the stop pin to protrude.

[0010] According to another embodiment, a system for tensioning a spinal system may include a flexible band configured to loop around a bone, a spinal rod configured for stabilizing two vertebrae, a band clamp implant having a recess for retaining the spinal rod and an opening for receiving the flexible band, and a tensioner instrument including a clip inserter and a ratcheting tensioner configured for tensioning the flexible band. The clip inserter may include a cannulated body and a collar configured to translate along the body and engage the spinal rod in a downward position. The ratcheting tensioner may include a fixed handle coupled to the clip inserter, a pivoting handle coupled to the fixed handle, a ratchet assembly including a ratchet and a spool keyed to the ratchet, and an actuator configured to engage the ratchet. When the pivoting and fixed handles are squeezed together, the actuator may force the ratchet and spool to rotate, thereby applying tension to the flexible band.

[0011] According to yet another embodiment, a method for tensioning a spinal system, in situ, may include one or more of the following steps in any suitable order: (1) looping a flexible band around a portion of bone, such as a lamina or transverse process; (2) threading the band through an implant; (3) affixing the implant to the clip inserter; (4) positioning the implant against a spinal rod and securing the clip inserter to the spinal rod; (5) threading the band through the spool; (6) positioning the tensioner against the implant; (7) squeezing the handles of the tensioner together causing the spool to rotate and wrapping the band around the spool, thereby applying controlled, incremental tension to the band; (8) once the desired tension has been reached, securing the band in the implant (e.g., with a set screw); and (9) removing the instrument from the patient.

[0012] Also provided are kits including implants of varying types and sizes, rods, tensioner

instruments of varying types and configurations, insertion tools, and other components for performing the procedure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0013] A more complete understanding of the present invention, and the attendant advantages and features thereof, will be more readily understood by reference to the following detailed description when considered in conjunction with the accompanying drawings, wherein:

[0014] FIG. 1 shows a perspective view of a handled tensioner instrument able to pinch and pull the flexible band to provide tension according to one embodiment;

[0015] FIG. 2 shows a perspective view of the tensioner instrument of FIG. 1 in a closed or relaxed position;

[0016] FIG. 3 shows a perspective view of the tensioner instrument of FIG. 1 in an open or tensioned position;

[0017] FIG. 4 is a close-up cross-sectional view of the distal tip of the tensioner instrument in the relaxed position;

[0018] FIG. 5 is a close-up cross-sectional view of the distal tip of the tensioner instrument in the tensioned position;

[0019] FIG. 6 shows a perspective view of an implant with the tensioner instrument in the relaxed position retaining the band;

[0020] FIG. 7 shows a perspective view of the implant with the tensioner instrument in the tensioned position, thereby applying a force to the band;

[0021] FIG. 8 shows a perspective view of a tensioner instrument system including a clip inserter and a ratcheting tensioner with a band extending therethrough according to one embodiment;

[0022] FIG. 9 shows a perspective view of the clip inserter of FIG. 8;

[0023] FIG. 10 shows a perspective view of the clip inserter engaging a band clamp implant;

[0024] FIG. 11 shows a perspective view of the clip inserter and band clamp engaged with the spinal rod;

[0025] FIG. 12 shows a side perspective view of the clip inserter, band clamp implant, and spinal rod with a driver shaft configured to engage with a set screw on the top of the band clamp;

[0026] FIG. 13 is a partial bottom view of the clip inserter showing a wire cut and collar pins in release grooves the body of the clip inserter;

[0027] FIG. 14 is a close-up cross-sectional view of the button in the clip inserter, at rest, with the threaded shaft engaged with the button;

[0028] FIG. 15 is a close-up cross-sectional view of the button depressed in the clip inserter and the threaded shaft disengaged;

[0029] FIG. 16 shows a perspective view of the ratcheting tensioner of FIG. 8;

[0030] FIG. 17 is cross-sectional view of the ratcheting tensioner;

[0031] FIG. 18 is an exploded view of the ratcheting tensioner;

[0032] FIG. 19 shows a perspective view of the ratcheting tensioner mated with the clip inserter;

[0033] FIG. 20 is a cross-sectional view of the ratcheting tensioner and clip inserter showing the button depressed and the stop pin disengaged from the clip inserter;

[0034] FIG. 21 is a cross-sectional view of the ratcheting tensioner and clip inserter showing the button at rest and the stop pin engaged with the clip inserter;

[0035] FIG. 22 shows a flexible band fed through the band clamp implant attached to the clip inserter and fed through the spool of the ratcheting tensioner;

[0036] FIG. 23 is a partial cross-sectional view showing the flexible band wrapped around the spool of the ratcheting tensioner;

[0037] FIG. **24** shows a perspective view of an overhead ratcheting tensioner according to one embodiment; and

[0038] FIG. **25** shows a system of tensioning a spinal construct using the grip style ratcheting tensioner of FIG. **8** and the overhead style ratcheting tensioner of FIG. **24**.

DETAILED DESCRIPTION OF THE INVENTION

[0039] Embodiments of the disclosure are generally directed to instrument, systems, and methods for tensioning flexible bands in-situ. Specifically, embodiments are directed to instruments and systems configured to tension a flexible band in order to provide fixation to the spine.

[0040] Additional aspects, advantages and/or other features of example embodiments of the invention will become apparent in view of the following detailed description. It should be apparent to those skilled in the art that the described embodiments provided herein are merely exemplary and illustrative and not limiting. Numerous embodiments of modifications thereof are contemplated as falling within the scope of this disclosure and equivalents thereto.

[0041] Referring now to FIGS. **1-7**, a handled tensioner instrument **10** is shown according to one embodiment. The tensioner **10** is configured to apply tension to an elongate member, cable, tether, cord, or band **12**. The band **12** may be a flexible member configured to be wrapped around bony anatomy or a portion of the spine, such as the lamina or transverse process, for example. Although described with reference to the spine, it will be appreciated that the instruments and systems described herein may be applied to other orthopedic locations and applications, such as trauma.

[0042] The flexible band **12** may be able to adapt to complex anatomies, such as severe spinal deformities. The band **12** may be used alone or in conjunction with an implant **14**, such as a screw (e.g., a pedicle screw), a clamp, a hook, or other suitable implant. The implant **14** may engage with an elongate member, such as a spinal rod, to provide fixation between vertebrae. In this embodiment, the implant **14** is shown as a pedicle screw with a tulip for receiving a spinal rod and a clamp for retaining the band **12**. Examples of other implants and rod constructs are described in more detail, for example, in U.S. Pat. Nos. 9,433,441; 10,034,692; 10,548,644; and 10,575,879, which are incorporated by reference herein in their entireties for all purposes. After the implant **14** is affixed to bone and/or secured to bone by looping the band **12** around the bony anatomy, correction of the spinal deformity may be achieved and held by the application of tension to the flexible band **12**.

[0043] The band **12** may be comprised of polyethylene terephthalates (PET), polyethylenes (e.g., ultrahigh molecular weight polyethylene or UHMWPE), polypropylenes, silk, polyamides, polyesters, polyacrylonitriles, silk cottons, combinations thereof, or other suitable biocompatible materials. The band **12** may be generally round, oval, or flat/tape geometry. The band **12** may transition from one geometry to another (e.g., a round to flat geometry or vice versa). If desired, the band **12** may be fully radiolucent or may have one or more marker strands that are designed to show up on fluoroscopy.

[0044] With emphasis on FIG. **1**, the tensioner **10** pinches and pulls the flexible band **12** in order to provide tension. The tensioner **10** includes an upper handle arm or first pivoting arm **20**, a lower handle arm or second pivoting arm **22**, and a base **24**. The first pivoting arm **20** extends from a first end or distal end **26** to a second end or proximal end **28**. Similarly, the second pivoting arm **22** extends from a distal end **30** to a proximal end **32**. The base **24** and distal ends **26**, **30** form the tip of the instrument **10**, which is configured to access the patient in-situ. The proximal ends **28**, **32** are manipulable by a user, such as a surgeon. The first and second pivoting arms **20**, **22** may each define a handle **60**, **62**, for example, near the proximal ends **28**, **32**, which are configured to be gripped and squeezed by the user. The inner facing portions of the handles **60**, **62** may include curved leaf springs **64**, **66** configured to apply an opposing force to the handles **60**, **62** when squeezed.

[0045] As best seen in FIGS. **6** and **7**, the base **24** may have a generally L-shaped body. For example, the base **24** may have a first elongate portion **34** and a second elongate portion **36** angled

relative to the first elongate portion **34**. A distal portion or first end **38** of the first elongate portion **34** of the base **24** may define a channel or slot **40** configured to retain the band **12** and/or guide the distal end **26** of the first arm **20**. The second end **42** of the first elongate portion **34** of the base **24** integrally connects to the second elongate portion **36**. The corner or second end **42** of the first elongate portion **34** of the base **24** may mate with the first arm **20**. The second elongate portion **36** of the base **24** terminates at a proximal end or free end **44**, which may be configured to mate with the second arm **22**.

[0046] As shown in FIGS. **2** and **3**, the instrument **10** may include a ratchet **52** positionable between the proximal ends **28**, **32** of the upper and lower pivoting arms **20**, **22** to hold tension applied to the flexible band **12**. The ratchet **52** may include a linear body or rail with a plurality of teeth **54** defined along the interior of the ratchet **52**. A first end of the ratchet **52** may be coupled to one of the pivot arms **20**, **22** via a pivot pin **56** and the opposite end of the ratchet **52** may be positionable through a slot or opening **58** in the body of the other pivot arm **20**, **22**. In FIG. **2**, the tensioner **10** is shown in a first relaxed position where tension would not be applied to the band **12**. In FIG. **3**, the tensioner **10** is shown in a second tensioned position where tension would be applied to the band **12**. As the ratchet **52** moves through the opening **58**, a pawl in the opening **58** engages the teeth **54** to thereby incrementally maintain the position of the arms **20**, **22** and the amount of tension applied to the band **12**.

[0047] With emphasis on FIGS. **4** and **5**, the base **24** may be fixed to each pivoting arm **20**, **22** to create two hinges **46**, **48**. The first hinge **46** connects the base **24** to the first arm **20**, for example, at the corner **42** connecting the first and second elongate portions **34**, **36** of the base **24**. The second hinge **48** connects the base **24** to the second arm **22**, for example, at the free end **44** of the second elongate portion **36** of the base **24**. The hinges **46**, **48** may each include a pivot pin. Although pivot pins are exemplified in this embodiment, it will be appreciated that other suitable joints could be selected.

[0048] The first hinge **46** moves the upper pivoting arm **20** toward the base **24**, while the second hinge **48** moves the base **24** away from the lower pivoting arm **22**. The distal end **26** of the upper pivoting arm **20** may include geometry configured to move within the slot **40** in the base **26**. In the rest/relaxed position shown in FIG. **4**, there is clearance between the distal end **26** of the upper pivoting arm **20** and the base **26** (slot **40**). When the handles **60**, **62** are squeezed into the tensioned position as shown in FIG. **5**, the upper pivoting arm **20** moves toward the base **24**, decreasing the clearance between the upper pivoting arm **20** and the base **26** until the upper pivoting arm **20** contacts the base **24** and travels along slot **40**. In this manner, the band **12** residing in slot **40** is pinched between the distal end **26** of the arm **20** and the base **24**.

[0049] As best seen in FIG. **3**, the lower pivoting arm **22** has a thru hole **50** configured to accept the flexible band **12**. The thru hole **50** is located directly beneath the slot **40** in the base **26**, thus allowing the flexible band **12** to be passed through the hole **50** in the lower pivoting arm **22** and into the slot **40** in the base **26**. When the handles **60**, **62** are squeezed, the upper pivoting arm **20** pinches the flexible band **12** against the base **26** and the lower pivoting arm **22** pushes the contacting surface away from the base **26**, thus providing tension to the flexible band **12**.

[0050] As seen in FIGS. **6** and **7**, the lower pivoting arm **22** may be placed against the implant **14** in order to tension the flexible band **12** through the implant **14**. As the handles **60**, **62** are compressed, the distance between the distal ends **26**, **30** of the arms **20**, **22** is increased, and a tension is applied to the band **12**. The ratchet **52** between the upper and lower pivoting arms **20**, **22** holds and maintains the tension applied to the band **12**.

[0051] During the procedure, the implant **14** may be secured to bone. In this embodiment, the implant **14** includes a pedicle screw with a tulip for receiving a spinal rod and an attached clamp for receiving the band **12**. It will be appreciated that other suitable implants may be selected. The band **12** may be threaded through the tensioner **12** and the implant **14**. The band **12** may be looped around a portion of bone, such as the lamina. A tension may be applied to the band **12** by squeezing

the handles **60**, **62** of the tensioner **12** toward one another. The ratchet **52** allows for controlled, incremental tensioning of the band **12**. Once the desired tension has been reached, the band **14** may be secured by the implant **12** (e.g., with a set screw in the clamp), and the instrument **10** may be removed from the patient.

[0052] Turning now to FIGS. **8-23**, an embodiment of a tensioner instrument system **100** is shown. The tensioner system **100** includes a clip inserter **102** and a ratcheting tensioner **104**, which are configured to mate together and apply tension to the band **12**. Although the tensioner system **100** is described with respect to two separate components, it will be appreciated that the instrument **100** may comprise a single integral body. The tensioner system **100** is configured to insert an implant **106** onto a spinal rod **108** and tension the flexible band **12** through the implant **106**.

[0053] As best seen in FIG. **10**, the implant **106** may be a band clamp implant having a generally c-shaped body with a recess **110** for retaining the rod **108**, one or more openings **112** for receiving the band **12**, and a locking member or set screw **114** for retaining the band **12** in the implant **106** and maintaining tension applied to the band **12**. Although the band clamp implant **106** is exemplified in this embodiment, it will be appreciated that the tensioner system **100** is configured to interface with other implants and apply tension to the band **12**. Examples of additional implants and rod constructs are described in more detail, for example, in U.S. Pat. Nos. 9,433,441; 10,034,692; 10,548,644; and 10,575,879, which are incorporated by reference herein in their entireties for all purposes.

[0054] Turning now to FIGS. **9-15**, the clip inserter **102** includes a main body **120**, a collar **122**, and a threaded shaft **124** extending through the main body **120**. With emphasis on FIG. **9**, the clip inserter **102** has a main body **120** extending from a tip or distal end **126** to a proximal end **128**. The main body **120** may be in the form of a hollow outer tube or cannula defining a central channel **118** between its distal and proximal ends **126**, **128**. The body **120** defines a recess **130** at the distal tip **126** which is configured to accept the band clamp implant **106**. The recess **130** may define one or more protrusions **132** configured to engage with a corresponding mating groove on the band clamp implant **106**.

[0055] As shown in FIG. **10**, the implant **106** is receivable in the recess **130** and mates with the protrusions **132** such that the implant **106** is retained at the distal end **126** of the main body **120** of the clip inserter **102**. It will be appreciated that the implant **106** may be coupled to the tip **126** of the clip inserter **102** in any suitable manner. As shown in FIG. **13**, a bottom surface of the main body **120** toward its distal end **126** may include a wire cut **132**. The wire cut **132** may extend along a longitudinal axis of the body **120** from the distal end **126** inward a distance past the collar **122** of the clip inserter **102**. The wire cut **132** allows the clip inserter **102** to splay open to accept and release the implant **106**.

[0056] The clip inserter instrument **102** has a collar **122** that is able to translate up and down the instrument **102** to engage and disengage with the spinal rod **108**. In the upward position, the collar **122** is disengaged from the rod **108**. In the downward position, the collar **122** is engaged with the rod **108**. The collar **122** may be a ring with a hollow center such that the main body **120** is received through the collar **122**. Toward the distal end **126** of the body **120**, the collar **122** may include one or more notches **134** configured to engage the spinal rod **108** when the collar **122** is in the downward position (as shown in FIG. **12**). The collar **122** may include a pair of spaced apart arms **136** which define the notches **134** at the free ends of the arms **136**. When the collar **122** moves distally, the notches **134** engage with the outer surface of the rod **108**, thereby securing the rod **108** to the clip inserter **102**.

[0057] The collar **122** is able to translate up and down the instrument **102** via engagement with the threaded shaft **124**. The threaded shaft **124** is receivable through the channel **118** in the main body **120**. The threaded shaft **124** may define a hollow body such that a driver shaft **138** is passable through the center of the threaded shaft **124**. The threaded shaft **124** includes a threaded portion **140** along the length of the shaft **124**. The threaded portion **140** may have one or more threads of

suitable diameter, handedness, thread form, thread angle, lead, pitch, etc. The threaded shaft **124** and collar **122** may cooperate as a ball screw, leadscrew, or other suitable translation mechanism. The proximal end of the shaft **124** may include a handle interface **142** (e.g., a ribbed portion) configured to mate with a handle (not shown). When the threaded shaft **124** is rotated, the collar **122** is translated upward to the disengaged position or downward to the engagement position with the rod **108**.

[0058] The threaded shaft **124** may be rotated in a first direction to translate the collar **122** downward. Actuation of the threaded shaft **124** pushes the collar **122** into contact with the spinal rod **108**, which is contacted on the underside by the band clamp implant **106**. These contact forces allow the band clamp implant **106** to be rigidly held in place on the spinal rod **108**. The threaded shaft **124** is cannulated to accept the driver shaft **138**, which is configured for tightening the set screw **114** on the band clamp **106**. The driver shaft **138** may also include a handle interface configured to mate with a handle (not shown).

[0059] As visible in FIG. 13, the main body **120** of the clip inserter **102** may include one or more grooves **144** configured to engage with pins **146** on the collar **122**. In particular, a pair of angled grooves **144** may be positioned on either side of the central wire cut **132**. The grooves **144** in the main body **120** may be angled inward toward the proximal end **128** and outward toward the distal end **126**. The angled grooves **144** allow the pins **146** in the collar **122** to force the distal end **126** of the body **120** to splay open when the collar **122** is pulled into the upward position, thereby releasing the implant **106** from the clip inserter **102**.

[0060] With emphasis on FIGS. 14 and 15, the clip inserter **102** may include a button **150** which is positioned within a slot **152** within the main body **120**. An inner portion of the button **150** may define a clearance hole **156** and threads **154** configured to engage the threaded shaft **124**. The button **150** may be held in place by a pin and contacts one or more springs **158**. In the rest position shown in FIG. 14, the threads **154** inside the button **150** engage the threaded portion **140** of the threaded shaft **124**. When the button **150** is depressed as shown in FIG. 15, the clearance hole **156** allows the threaded shaft **124** to translate freely through the button **150**.

[0061] Turning now to FIGS. 16-23, the tensioner system **100** includes a ratcheting tensioner **104**. The ratcheting tensioner **104** is a handled instrument which uses a ratchet assembly **220** to rotate a spool **222** in order to wind up the band **12**, thereby providing tension to the band **12**. An exploded view of the ratcheting tensioner **104** is shown in FIG. 18. The ratcheting tensioner **104** may include a fixed handle **170**, a pivoting handle **172**, and a ratchet assembly **220** with an actuator assembly **240** and a release assembly **254**.

[0062] With emphasis on FIG. 16, the ratcheting tensioner **104** has a fixed handle **170** and a pivoting handle **172**. The fixed handle **170** extends from a distal end **174**, which is configured to couple to the clip inserter **102**, to a proximal end **176**. The proximal end **176** includes a handle portion **178** configured to be manipulated by the user. A neck **180** connects the distal portion **174** to the handle portion **178**. The neck **180** may be bent or angled upwards to provide a pistol grip orientation for the ratcheting tensioner **104** when assembled to the clip inserter **102** and used to tension the band **12**. For example, the neck **180** may be curved back toward the fixed handle **170**. The distal portion **174** of the fixed handle **170** may define an opening or through slot **182** configured to receive the main body **120** of the clip inserter **102**. The distal portion **174** may include a ring or loop **184** defining the slot **182** for retaining the clip inserter **102**. For example, the body **120** of the clip inserter **102** may be inserted through the slot **182** in the fixed handle **170** near the proximal end **128** of the clip inserter **102**.

[0063] As shown in FIGS. 17 and 18, the ratcheting tensioner **104** may be coupled to the clip inserter **102** with a button assembly **190**. The button assembly **190** may include a button **192**, a button pin **194** for securing the button **192**, a stop pin **196** for engaging with the body **120** of the clip inserter **102**, and a button spring **198** causing the stop pin **196** to protrude into the slot **182**. Intersecting the slot **182** is a bore **202** sized and dimensioned to accept the button spring **198** and

the stop pin **196**. At rest, the button spring **198** causes the stop pin **196** to protrude into the slot **182** which accepts the clip inserter **102**. The fixed handle **170** has a first through hole **204** sized and dimensioned to accept the button **192** and a second transverse through hole **206** configured to accept the button pin **194**. The button **192** is coupled to the stop pin **196** with the button pin **194** such that the stop pin **196** translates back into the fixed handle **170** when the button **192** is depressed. When the button **192** is depressed as shown in FIG. **20**, the stop pin **196** retracts and the fixed handle **170** may be engaged onto the clip inserter **102**. When the button **192** is released as shown in FIG. **21**, the stop pin **196** translates and extends outwardly into a mating hole **208** on the clip inserter **102** to securely couple the ratcheting tensioner **104** to the clip inserter **102**.

[0064] The pivotable handle **172** extends from a distal end **210**, which couples to the fixed handle **170**, to a proximal end **212**. The proximal end **212** includes a handle portion **214** configured to be manipulated by the user. The distal end **210** is pivotably coupled to the fixed handle **170** with a ratchet assembly **220**. The pivotable handle **172** and fixed handle **170** are sized and dimensioned to retain the ratchet assembly **220**.

[0065] The ratchet assembly **220** includes a spool **222** and at least one rotary ratchet **224**. For example, the assembly **220** may include first and second rotary ratchets **224** positionable on opposite sides of the spool **222**. The spool **222** includes two half rounds or halves **232** with a space or gap **230** separating the two halves **232**. The outer surfaces of the halves **232** may be curved or rounded. Each rotary ratchet **224** may include a round gear having a plurality of teeth **234** around the perimeter of the ratchet **224**. The teeth **234** may be uniformly distributed around the body of the ratchet **224**. The actuator **242** is configured to engage the teeth **234** as the ratchets **224** rotate. The teeth **234** may be sloped or angled to allow the actuator **242** to slide up and over the teeth **234** and into the depression between teeth **234** when the ratchets **224** rotate in a first direction. When the ratchets **224** try to move in an opposite direction, the actuator **242** catches against the first tooth **234**, thereby locking the actuator **242** against the tooth **234** and preventing any further motion in that direction. The ratchet assembly **220** is secured to the handles **170**, **172** with first and second cap plates **236** positioned on the outer sides of the ratchets **224**. A plurality of fasteners **238**, such as set screws, may be used to secure the cap plates **236** and the entire assembly.

[0066] Each handle **170**, **172** has a thru hole **226** sized and dimensioned to accept the spool **222**. The two ratchets **224** sit between the pivoting handle **172** and the fixed handle **170** on either side of the fixed handle **170**. Each ratchet **224** defines two cutouts **228** configured to accept the two halves of the spool **222**. The spool **222** is able to rotate independently of the handles **170**, **172**, but the spool **222** is keyed to the ratchet **224** such that the spool **222** rotates when the ratchet **224** rotates. The space or gap **230** between the two halves of the spool **222** is large enough to accept the flexible band **12**.

[0067] The actuator assembly **240** includes an actuator **242** for engaging with the ratchets **224**, an actuator button **244** for engaging and disengaging the actuator **242**, an actuator pin **246** for securing the actuator **242**, and an actuator spring **248** causing the actuator **242** to contact the ratchets **224**. The actuator **242** includes a body with two spaced apart pawls **250**. Each pawl **250** is configured to engage with teeth **234** on the respective ratchets **224**. The pivoting handle **172** has a bore to accept the actuator **242** and actuator spring **248**. The actuator spring **248** forces the actuator **242** to contact the ratchets **224** in the resting position. The pivoting handle **172** has two transverse thru holes to accept the actuator button **244** and the actuator button pin **246**. The actuator button **244** is coupled to the actuator **242** with the actuator pin **246** such that when the button **244** is depressed, the actuator **242** translates away from the ratchets **224**, thereby disengaging the actuator **242** from the ratchets **224**.

[0068] The release assembly **254** includes a release arm **256**, a release spring **258**, and a release pin **260** for pivotably securing the release arm **256**. The release arm **256** include a body with a thumb press **262** and a pair of spaced apart tongues **264** for engaging the ratchets **224**. The release arm **256** is pivotably coupled to the fixed handle **170**. The release arm **256** is positioned within a slot

and is secured by pivot pin **260**. The release arm **256** defines a counter bore to accept the release spring **258**. The release spring **258** forces the tongues **264** of the release arm **256** into contact with the ratchets **224** at rest. The tongues **264** of the release arm **256** may be disengaged from the ratchets **224** by depressing the release arm **256**, thereby compressing the release spring **258** and pivoting the tongues **264** out of contact with the ratchets **224**.

[0069] The ratcheting tensioner **104** may include a pair of leaf springs **266** coupled to the handles **170**, **172** with one or more fasteners **268**, such as a plurality of set screws. The curved leaf springs **266** cause the handles **170**, **172** to remain open at rest. When the handles **170**, **172** are squeezed together, the actuator **242** on the pivoting handle **172** contacts the ratchets **224** and forces the ratchets **224** and spool **222** to rotate. While the ratchet **224** is rotating, the release arm **256** on the fixed handle **170** is lifted into consecutive ratchet positions. When the handles **170**, **172** are released, the leaf springs **266** cause the handles **170**, **172** to open and the actuator **242** on the pivoting handle **172** lifts back into consecutive ratchet positions. The spool **222** is held in the new orientation and does not rotate back with the opening pivoting handle **172** because the release arm **256** on the fixed handle **170** is engaged with the ratchet **224**. This process is repeated causing the spool **222** to rotate within the ratcheting tensioner **104**, thereby applying tension to the band **12**.

[0070] As shown in FIGS. **22-23**, the ratcheting tensioner **104** is engaged with the clip inserter **102** with button assembly **190**. The flexible band **12** is fed thru the band clamp implant **106** and the band clamp **106** is placed onto the rod **108** with the collar **122** of the clip inserter **102**. The flexible band **12** is fed between the two halves **232** of the spool **222** of the ratcheting assembly **220**. The handles **170**, **172** of the ratcheting tensioner **104** are then actuated to wrap the flexible band **12** around the spool **222**, thereby pulling the flexible band **12** thru the band clamp **106** and applying tension. Once a sufficient amount of tension is achieved, the driver shaft **138** may be passed through the clip inserter **102** into the set screw **114** on the band clamp **106** to lock the tension in the band **12**. Tension may be removed from the flexible band **12** by unwinding the spool **222**. In order to do this, the release arm **256** and actuator button **244** are depressed to disengage the ratchets **224** and the spool **222** is rotated manually to unwind the spool **222**.

[0071] During the procedure, the band **12** may be threaded through the implant **106** and through the gap **230** in the spool **222** of the ratcheting tensioner **104**. The band **12** may be looped around a portion of bone, such as the lamina or transverse process. Tension may be applied to the band **12** by squeezing the handles **170**, **172** of the ratcheting tensioner **104**. As the spool **222** rotates, the band **12** wraps around the spool **222** applying controlled, incremental tension to the band **12**. Once the desired tension has been reached, the band **12** may be secured by the implant **106** (e.g., with a set screw **114** in the clamp **106**), and the instrument **100** may be removed from the patient.

[0072] Turning now to FIGS. **24** and **25**, an alternative version of a tensioner instrument system **300** is shown, which is the same as tensioner **100**, except with a different overhead style grip configuration. Similar to the pistol grip style tensioner system **100**, the overhead style tensioner system **300** includes the clip inserter **102** and ratcheting tensioner **104**. In tensioner system **300**, however, the ratcheting tensioner **104** is configured to be placed in an overhead orientation with respect to the clip inserter **102**. For the overhead system **300**, the ratcheting tensioner **104** is configured to be placed in the overhead orientation because the neck **302** of the fixed handle **170** is angled opposite of neck **180**. For example, the neck **302** of tensioner system **300** is angled or curved back toward the pivoting handle **172**.

[0073] As shown in FIG. **25**, the overhead style tensioner system **300** offers the surgeon another option depending on their preference of hand position during tensioning of the flexible band **12**. If desired, the overhead style tensioner system **300** may be used in conjunction with the pistol grip style tensioner **100**. Band clamps **106** may be used at sequential spinal levels in order to reduce a deformity and anatomy may dictate that the band clamps **106** are positioned within close proximity to one another. Therefore, it may be advantageous to offer both styles of ratcheting tensioner **100**, **300** in order to give space for the surgeon's hands.

[0074] The systems described herein allow surgeons to tension the flexible band in order to correct spinal deformities and achieve fixation. The instruments offer easy engagement with the band clamps and flexible bands, which may save time, for example, compared to instruments that use secondary locking steps for the band clamps or flexible bands. The tensioning instruments allow for essentially limitless tensioning capacity, which may be an improvement over instruments limited by the travel range of threaded mechanisms, for example. The tensioners may save the surgeon time during surgery as the tensioner does not need to be reset during correction. In addition, the handle mechanisms may allow for ease of use without the need for additional actuating instruments. The variety of configurations may also allow the surgeon to customize the implementation of instruments to the patient in order to optimize visualization of the surgical site. [0075] Although the invention has been described in detail and with reference to specific embodiments, it will be apparent to one skilled in the art that various changes and modifications can be made without departing from the spirit and scope of the invention. Thus, it is intended that the invention covers the modifications and variations of this invention provided they come within the scope of the appended claims and their equivalents. It is expressly intended, for example, that all components of the various devices disclosed above may be combined or modified in any suitable configuration.

Claims

1. A tensioner instrument as substantially described herein.
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